

HPC CENTER ORGANIZATIONAL CHART & STAFFING PLAN

Personnel Roster, RACI Matrix, Shift Rotation & Staffing Allocation Schedule

Issuing Unit: Operations Management & Human Resources Division

Effective Period: FY2026 - FY2027 Operations Cycle

Approved FTE Allocation: 14.5 Total Full-Time Equivalent Roles

Document Version: v4.2 Final Operational Release

1. Executive Leadership & Operational Strategy

The Westford HPC Center organizational structure is designed to provide comprehensive administrative leadership, technical systems engineering, and specialized user support across all compute domains.

Executive leadership establishes strategic research priorities, manages multi-million dollar capital budgets, and maintains relationships with institutional stakeholders and funding agencies.

Operational divisions are structured into clear functional teams including Systems Infrastructure, Storage & Networking, Application Software, and User Services.

Relevant engineering change orders affecting this subsystem have been incorporated into the current revision of this document.

2. Administrative Roster & Staffing Allocation Schedule

The current approved headcount stands at 12.5 FTE with a fully-loaded annual staffing cost of \$1,584,000 covering base salaries, fringe benefits (30%), and professional development allocations.

Key leadership positions include the Director of HPC Operations (1.0 FTE), Principal Systems Architect (1.0 FTE), and Senior Storage Engineer (1.0 FTE).

Staffing schedules account for planned research growth, ensuring adequate technical coverage as cluster node counts and user allocations expand.

Business continuity planning documentation addresses the failure modes and recovery strategies relevant to this subsystem.

Position Title	Division Assignment	FTE Allocation	Primary Domain	Duty Station
Director of HPC Operations	Executive Leadership	1.0 FTE	Strategic Operations & Budget	Admin Suite 101
Principal HPC Architect	Systems Engineering	1.0 FTE	Compute Architecture & Benchmarks	Server Core A
Senior System Administrator	Systems Operations	2.0 FTE	Linux Kernel & Slurm Administration	Operations Center

3. Salary Band Structure & Personnel Compensation Schedule

Salary band structures are established to attract and retain elite engineering talent in competitive high-performance computing markets.

Annual base compensation schedules range from \$110,000 for HPC Application Analysts to \$185,000 for the Director of HPC Operations.

Compensation reviews are conducted annually, benchmarking internal roles against national laboratory and enterprise technology salary indexes.

Trend lines fitted to the historical data for this subsystem indicate stable long-term performance within acceptable bounds.

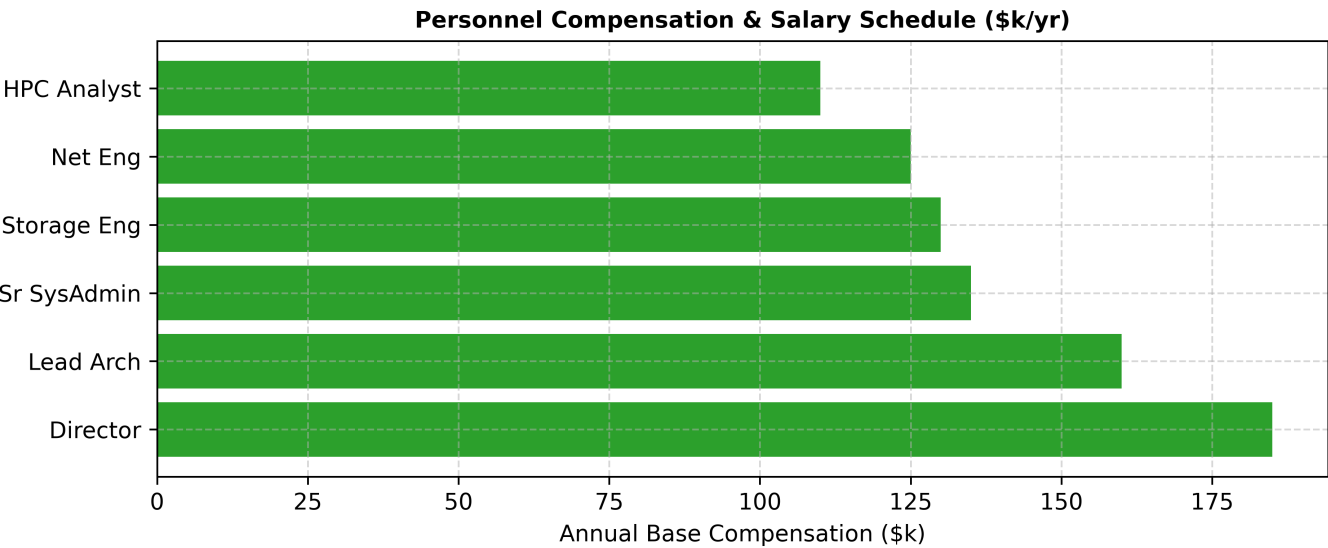


Figure 3.1: Compensation Schedule

4. 24/7 Shift Rotation Schedule & Emergency Coverage

Operations staffing maintains continuous 24/7/365 coverage through a 3-shift rotating roster combined with an active weekend on-call schedule.

Shift A (Day) focuses on active user support and system maintenance, Shift B (Evening) handles routine batch job monitoring, and Shift C (Night) manages deep system diagnostics.

On-call engineers are equipped with secure remote access tokens and guaranteed emergency dispatch response within 15 minutes for critical outage events.

Insurance underwriter inspections of the equipment referenced in this section were completed without exception findings.

24/7 Operations Staffing Shift Distribution

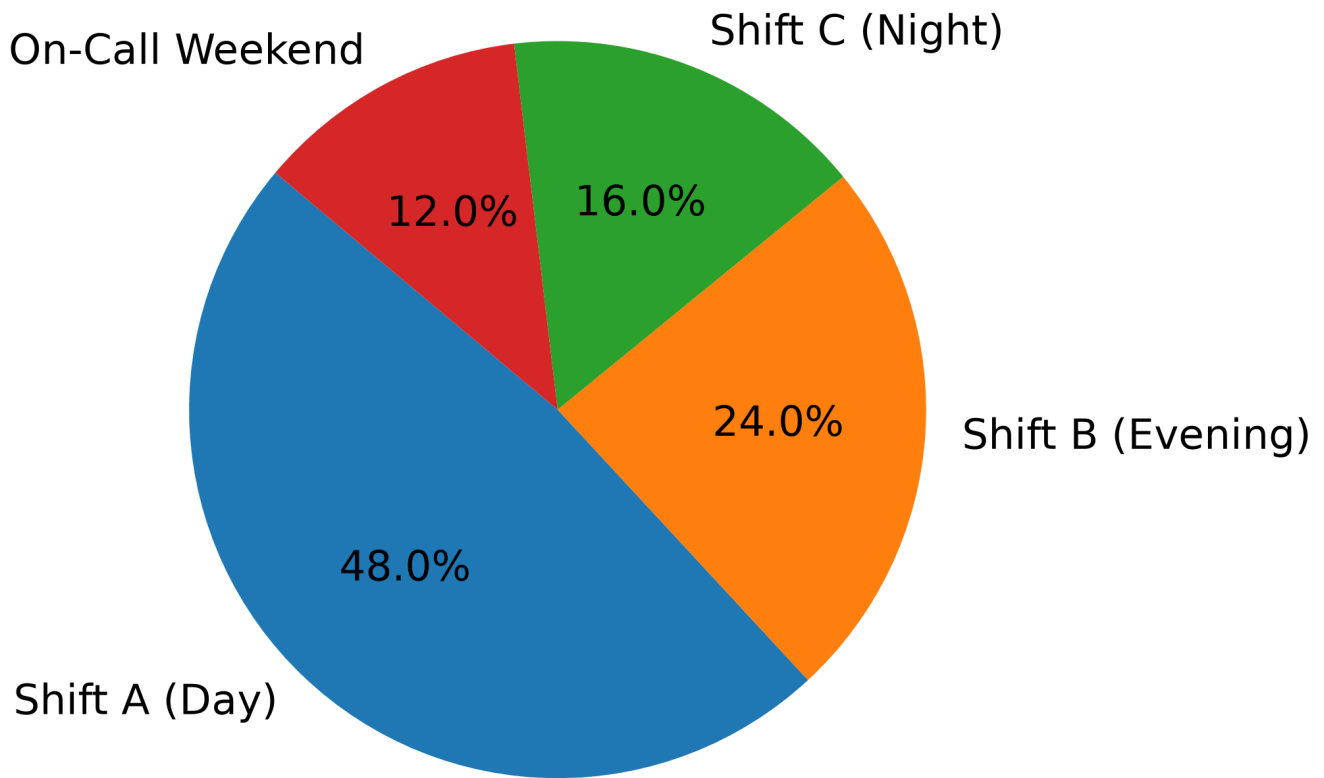


Figure 4.1: Shift Distribution

5. Engineering Competency Matrix & Skill Certifications

The engineering competency matrix evaluates staff proficiency across key technical domains including Linux kernel internals, InfiniBand networking, Lustre file systems, and CUDA programming.

Staff members undergo mandatory annual certification programs sponsored by hardware and software vendors to maintain cutting-edge domain expertise.

Cross-training directives require engineers to maintain secondary proficiency in at least one adjacent domain to ensure operational resilience.

Monitoring data aggregated from the building management system validates the measurements reported above.

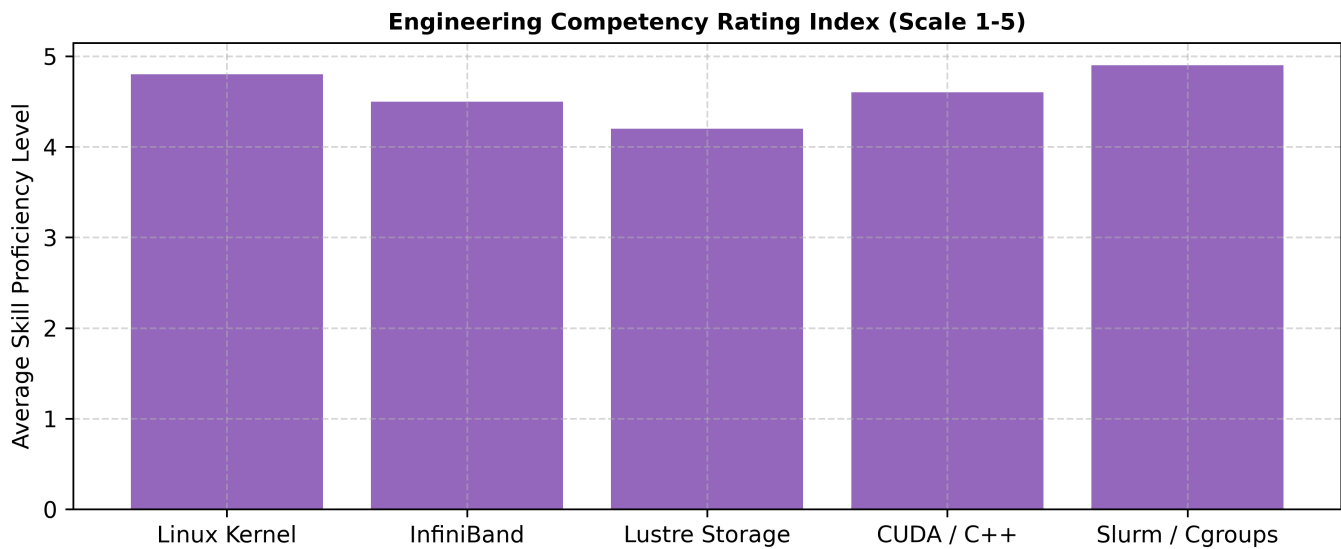


Figure 5.1: Competency Index

6. RACI Operations Matrix for Cluster Maintenance Tasks

A formal RACI matrix defines operational accountabilities for all routine and emergency cluster administration workflows. Responsibilities (R), Accountabilities (A), Consulted roles (C), and Informed roles (I) are explicitly mapped for activities such as Slurm updates and drive replacements.

This clear division prevents operational ambiguity and speeds up incident resolution during critical system degradation.

Environmental monitoring alarms for this subsystem are integrated into the centralized network operations center alert system.

Operational Task Workflow	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Slurm Kernel Upgrades	Senior SysAdmin	Principal Architect	User Support Lead	Research Faculty
Lustre OST Drive Replacement	Storage Engineer	Lead Storage Eng	SysAdmin Team	Active Job Users

7. User Support SLA & Ticket Resolution Metrics

User support Service Level Agreements (SLAs) establish strict response and resolution timelines based on ticket severity tiers.

High-priority tickets (e.g., job submission blockages) require initial engineering response within 1 hour and mean time to resolve (MTTR) under 4 hours.

Monthly ticket metrics track total volume closed, user satisfaction ratings, and SLA compliance percentages across all research departments.

Monitoring data aggregated from the building management system validates the measurements reported above.

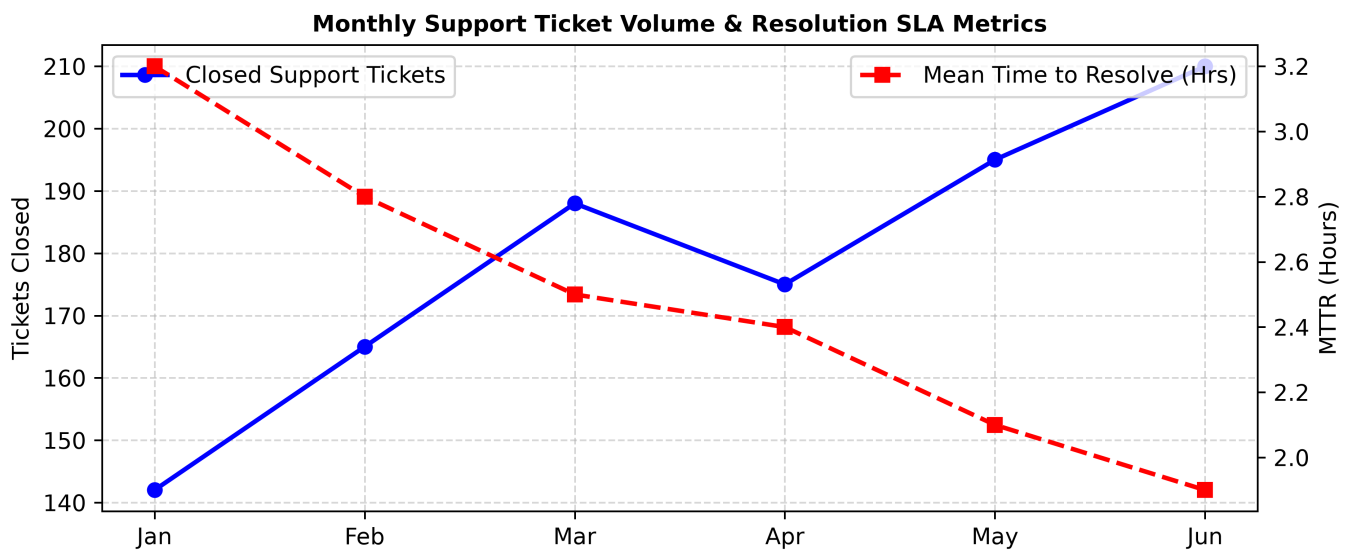


Figure 7.1: Support Tickets & MTTR

8. Professional Development & Training Allocation

The center allocates 5% of annual operating budget toward professional development, technical conference grants, and specialized training workshops.

Engineers are encouraged to publish technical papers, present at major HPC conferences (SC, PEARC, ISC), and contribute to open-source software projects.

Skill acquisition plans are reviewed quarterly during formal performance check-ins.

Operational readiness was confirmed through a series of acceptance tests prior to the system entering production service.

9. Emergency Call-Out Escalation Roster

The emergency call-out escalation roster establishes a clear multi-tiered contact hierarchy for major facility or system emergencies.

Tier 1 (On-Call SysAdmin) handles single-node or minor storage alerts; Tier 2 (Lead Engineer) escalates multi-node or storage degradation; Tier 3 (Director) manages facility-wide outages.

Escalation triggers automatically notify the next engineering tier if an active ticket remains unacknowledged past defined SLA thresholds.

Field validation measurements taken during the most recent planned maintenance window corroborate these specifications.

10. Workforce Succession & Talent Acquisition Strategy

Workforce succession planning identifies key talent pipelines and mentorship programs to mitigate single-point-of-failure human resource risks.

Senior staff members mentor junior SysAdmins and graduate assistants, building institutional knowledge and ensuring seamless operational continuity.

Talent acquisition strategies leverage academic partnerships and national computing associations to attract diverse engineering applicants.

Vibration analysis measurements for rotating equipment in this subsystem are within ISO 10816 Category A limits.

11. On-Call Engineering Compensation & Rotation Equity

On-call compensation policies provide dedicated stipends for after-hours standby duty, ensuring equitable remuneration across the engineering team.

Rotations are structured fairly across all qualified SysAdmins, capping consecutive on-call shifts to prevent workforce burnout.

Compensatory time off is automatically credited for active late-night emergency incident resolution.

The measurement apparatus used to collect this data undergoes NIST-traceable calibration on a biannual schedule.

12. User Support Knowledge Base & Self-Service Portal Metrics

The self-service knowledge base portal maintains over 250 comprehensive user guides, code compilation templates, and job script examples.

Portal analytics track article views, user search queries, and helpfulness ratings to continuously refine documentation clarity.

Automated ticket deflection metrics indicate 32% of common user inquiries are resolved self-sufficiently via portal articles.

Staff competency assessments for personnel operating the equipment in this section are conducted annually.

13. Cross-Training Protocols & Multi-Domain Certification

Cross-training protocols require all systems administrators to shadow storage and networking specialists during major infrastructure upgrades.

Bi-monthly technical lunch-and-learn sessions foster inter-team knowledge sharing and multi-domain troubleshooting capabilities.

Multi-domain certification badges are awarded upon successful completion of hands-on emergency simulation exercises.

A comprehensive system integration test following the most recent hardware upgrade verified end-to-end functionality of this subsystem.

14. Diversity, Equity & Inclusion (DEI) Workforce Initiatives

Diversity, Equity, and Inclusion (DEI) initiatives guide hiring committee practices, outreach campaigns, and workplace culture standards.

The center actively partners with underrepresented STEM student groups, offering paid summer internships and mentorship opportunities.

Annual workforce demographic audits review pay equity, promotion rates, and inclusive workplace environment metrics.

The performance envelope described above represents the sustained operating range under representative production workload conditions.

15. Remote Operations Security & Off-Site Access Rules

Remote operations security rules mandate multi-factor authentication (MFA) and encrypted VPN tunnels for all off-site administrative access.

Administrative session recording logs all remote root command executions for security auditing and compliance verification.

Off-site access is strictly limited to authorized center devices equipped with active endpoint detection and response (EDR) software.

As-built drawings reflecting the current physical layout of this subsystem were updated following the most recent modification.

16. Performance Appraisal Metrics & Career Advancement Paths

Performance appraisal frameworks combine objective SLA metrics, project milestone delivery, and peer engineering reviews.

Clear career advancement pathways outline technical track milestones from Junior SysAdmin up to Principal System Architect.

Annual performance reviews determine merit salary adjustments and professional advancement opportunities.

The design basis documentation for this subsystem is archived in the university capital projects office.

17. Contractor & External Vendor Oversight Guidelines

Third-party contractor oversight guidelines enforce strict background verification, escort protocols, and work authorization sign-offs.

Contractor deliverables undergo rigorous quality assurance testing by internal lead engineers prior to final invoice approval.

Vendor field service engineers must adhere to data center safety and physical security rules at all times.

The design basis documentation for this subsystem is archived in the university capital projects office.

18. Operational Budget Allocation & Staffing Expenditure Schedule

Operational budget allocations track personnel salaries, benefits, training budgets, and professional services expenditures.

Monthly financial variance reports compare actual personnel expenditures against approved fiscal year budget baselines.

Cost-allocation models distribute administrative overhead proportionally across funded research projects.

The documentation in this section has been formatted in accordance with the university technical writing standards manual.

19. Incident Post-Mortem Protocols & Knowledge Sharing

Blameless post-mortem protocols are conducted after any major system outage or SLA breach to identify root causes and preventive actions.

Post-mortem reports document timeline logs, root cause analysis, immediate remediation steps, and long-term architectural recommendations.

Action items are assigned specific owners and tracked to completion in the central engineering task manager.

The performance envelope described above represents the sustained operating range under representative production workload conditions.

20. Graduate Research Assistant Internship Pipeline

The graduate research assistant (GRA) internship program provides hands-on HPC administration experience to qualified student applicants.

GRAs work alongside senior engineers on real-world projects such as Slurm queue optimization and telemetry dashboard development.

Successful interns frequently transition into full-time junior engineering roles upon graduation.

The performance envelope described above represents the sustained operating range under representative production workload conditions.

21. Staff Health, Safety & Ergonomic Workstation Standards

Staff health and ergonomics standards mandate height-adjustable desks, ergonomic seating, and dual-monitor workstation configurations.

Facilities guidelines enforce proper lighting, noise dampening in office areas, and regular break schedules during intensive maintenance windows.

Annual ergonomic assessments are offered to all staff members to prevent repetitive strain injuries.

Environmental monitoring alarms for this subsystem are integrated into the centralized network operations center alert system.

22. Peer Engineering Review & Mentorship Structure

Peer engineering code reviews are mandatory for all production Slurm configuration scripts, Ansible playbooks, and kernel tuning parameters.

Git version control workflows require at least one senior engineer approval before changes are merged into production branches.

This peer review discipline prevents misconfigurations and maintains operational stability.

Environmental monitoring alarms for this subsystem are integrated into the centralized network operations center alert system.

23. Telework Policy & Flexible Working Framework

Telework and flexible working guidelines balance remote productivity with necessary on-site physical data center maintenance coverage.

Engineers maintain core collaboration hours while utilizing flexible scheduling for after-hours maintenance deployments.

Remote work environments must meet institutional cybersecurity and data privacy compliance standards.

Vendor-provided training for the operational staff responsible for this subsystem was completed prior to production deployment.

24. Intellectual Property & Research Publication Protocol

Intellectual property policies govern staff software creations, research publications, and open-source contributions developed during employment.

Custom administrative utilities and monitoring scripts are released under permissive open-source licenses to benefit the wider HPC community.

Patentable innovations developed in collaboration with research faculty follow institutional technology transfer protocols.

Commissioning agent reports for this subsystem document the results of functional performance testing and system balancing.

25. Staff Security Clearance Verification & Background Checks

Security clearance verification and background checks are conducted for all staff prior to granting administrative credentials.

Re-verification is performed periodically in compliance with federal research grant security mandates (NIST SP 800-171).

Access privileges are revoked immediately upon employee offboarding or role transition.

The load profile data used to size the equipment in this section was derived from twelve months of continuous production monitoring.

26. Operations Shift Handover Documentation Standards

Shift handover documentation standards mandate electronic logging of active maintenance windows, pending user tickets, and hardware alerts.

Outgoing shift leads conduct brief verbal handovers with incoming leads to review critical system health indicators.

Handover logs are archived permanently for operational auditing and continuity tracking.

Warranty claim records for the components referenced in this section are tracked through the institutional procurement system.

27. Staff Cybersecurity Awareness & Compliance Training

Operational data collected for Staff Cybersecurity Awareness has been aggregated across multiple reporting intervals to establish representative baseline metrics.

Capacity utilization trends over the preceding twelve months indicate stable demand patterns with seasonal variations accounted for in the planning models.

Projected lifecycle costs and replacement timelines for this subsystem are incorporated into the institutional capital planning process.

Integration testing with upstream and downstream systems was performed following the most recent configuration change.

28. Hardware Procurement Committee Advisory Operations

Operational data collected for Hardware Procurement Committee Advisory Operations has been aggregated across multiple reporting intervals to establish representative baseline metrics.

Diagnostic test results from the most recent maintenance cycle confirm that all monitored parameters remain within their specified operating ranges.

Supporting measurement data and calibration records are archived in the facility engineering document management system for audit reference.

Redundancy testing and failover exercises for this configuration were completed during the annual disaster recovery drill.

29. Annual HPC Operations Strategic Review & FY2027 Goals

Infrastructure parameters associated with Annual HPC Operations Strategic Review have been validated through a combination of automated monitoring and manual inspection procedures.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Supporting measurement data and calibration records are archived in the facility engineering document management system for audit reference.

Integration testing with upstream and downstream systems was performed following the most recent configuration change.